
Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

English I

BCA — Semester 1 — 2018 — Answer Sheet

Subject Code: CACS103

Group B

Attempt any SIX questions. [6×5=30]

11. What are the negative aspects of replacing humans with computers?

[5]

Answer: Negative Aspects of Replacing Humans with Computers:

- 1. Unemployment and Job Displacement:** Automation eliminates many manual and routine jobs. Factory workers, bank tellers, and data entry operators face unemployment as machines take over their roles. This creates economic instability for affected families and communities.
- 2. Loss of Human Touch and Empathy:** Computers cannot replicate genuine human emotions, empathy, and personal connection. In healthcare, education, and customer service, automated systems often provide cold, impersonal interactions that fail to address emotional needs.
- 3. Over-dependence and Skill Atrophy:** As people rely more on computers for calculations, navigation, and memory, basic human skills deteriorate. Students using calculators may struggle with mental arithmetic; GPS dependence reduces spatial awareness.
- 4. Privacy and Security Risks:** Computerized systems collect vast amounts of personal data, creating risks of identity theft, surveillance, and unauthorized access. A single data breach can affect millions of users.
- 5. Technical Failures:** When computer systems crash or malfunction, entire operations can halt. Unlike humans who can improvise, computers follow rigid programming and cannot adapt to unexpected situations without human intervention.
- 6. High Costs of Implementation:** Developing, installing, and maintaining computerized systems requires significant investment. Small businesses and developing countries may not afford such technology, widening the digital divide.
- 7. Ethical and Moral Limitations:** Computers cannot make ethical judgments involving context, compassion, and moral nuance. Automated decision-making in criminal justice, hiring, or healthcare can perpetuate biases embedded in algorithms.

12. Write a paragraph explaining how information is processed by information system.

[5]

Answer: Information Processing by Information Systems:

An information system processes data through a systematic cycle to produce meaningful information for decision-making. The process begins with **input**, where raw data is collected from various sources such as keyboards, scanners, sensors, or external databases. This raw data is then fed into the system for **processing**, which involves organizing, calculating, sorting, classifying, and summarizing the data using hardware and software resources. During processing, the system applies logical operations, mathematical calculations, and business rules to transform raw data into structured formats. The processed data undergoes **storage** in databases or file systems for future retrieval and reference. Finally, the system generates **output** in the form of reports, graphs, tables, or dashboards that present the information in a user-friendly manner. This output is then distributed to decision-makers who use it for planning, controlling, and managing organizational activities. The entire process is supported by feedback loops where the outcomes of decisions are monitored and used to refine future inputs, creating a continuous cycle of improvement. Effective information processing ensures accuracy, timeliness, and relevance of information, which are critical for organizational success in the modern digital economy.

13. How are the multi-user databases managed by a piece of software?

[5]

Answer: Management of Multi-User Databases by DBMS Software:

A Database Management System (DBMS) is specialized software that manages multi-user databases by coordinating access, ensuring data integrity, and maintaining security across concurrent users.

1. Concurrency Control:

The DBMS manages simultaneous access through techniques like:

- **Locking:** Prevents multiple users from modifying the same data simultaneously. Shared locks allow reading; exclusive locks allow writing.
- **Timestamp Ordering:** Assigns timestamps to transactions and ensures they execute in chronological order.
- **Optimistic/Pessimistic Concurrency Control:** Either assumes conflicts are rare (optimistic) or prevents them proactively (pessimistic).

2. Transaction Management:

Ensures database consistency through ACID properties:

- **Atomicity:** All operations in a transaction complete successfully or none do.
- **Consistency:** Database remains in a valid state before and after transactions.
- **Isolation:** Concurrent transactions do not interfere with each other.
- **Durability:** Committed changes survive system failures.

3. Security and Access Control:

- Authentication verifies user identities.
- Authorization grants specific permissions (read, write, delete) based on roles.
- Encryption protects sensitive data during storage and transmission.

4. Backup and Recovery:

Regular backups and log files enable restoration after crashes, hardware failures, or data corruption.

5. Query Optimization:

The DBMS analyzes queries and chooses the most efficient execution plan to minimize resource usage and response time.

14. Make a list of jobs suitable for robots, and those to be done only by humans.

[5]

Answer: Jobs Suitable for Robots vs. Jobs for Humans:

Jobs Suitable for Robots:

1. **Repetitive Manufacturing:** Assembly line work, welding, painting cars — robots perform consistently without fatigue.
2. **Hazardous Environments:** Bomb disposal, nuclear reactor inspection, deep-sea exploration, mining in dangerous conditions.
3. **Precision Tasks:** Microsurgery assistance, semiconductor fabrication, watchmaking — robots exceed human precision.
4. **Heavy Lifting:** Warehouse automation, construction material handling, cargo loading.
5. **Data Processing:** Sorting, categorizing, and analyzing large datasets; automated bookkeeping.
6. **Cleaning and Maintenance:** Floor cleaning robots, window washing on skyscrapers, pipe inspection.
7. **Space Exploration:** Mars rovers, satellite repairs, extraterrestrial sample collection.

Jobs to be Done Only by Humans:

1. **Creative Arts:** Writing novels, composing music, painting original artworks — requires imagination and emotional depth.
2. **Emotional Care:** Nursing, counseling, childcare, elderly care — requires empathy and genuine human connection.
3. **Complex Decision-Making:** Judges, diplomats, strategic planners — require wisdom, ethics, and contextual understanding.
4. **Education and Mentoring:** Teaching, coaching, inspiring students — requires adaptability and personal rapport.
5. **Ethical Leadership:** CEOs, politicians, religious leaders — require moral judgment and vision.
6. **Research and Innovation:** Scientists forming new hypotheses, entrepreneurs identifying unmet needs — requires curiosity and intuition.
7. **Social Work:** Community organizing, conflict resolution, humanitarian aid — requires cultural sensitivity and trust-building.

15. What do you mean by virtual reality? How is it going to affect young people's attitude to violence?

[5]

Answer: Virtual Reality (VR):

Virtual Reality is a computer-generated simulation of a three-dimensional environment that users can interact with in a seemingly real way using special electronic equipment such as headsets with screens, gloves with sensors, and motion controllers. VR creates an immersive experience that blocks out the physical world and replaces it with a digital one.

Effects on Young People's Attitude to Violence:

1. **Desensitization:** Repeated exposure to violent content in VR games and experiences can reduce emotional sensitivity to real-world violence. Young people may become numb to suffering and view aggression as normal.
2. **Behavioral Modeling:** VR provides highly realistic experiences where users physically enact violent behaviors. This active participation (unlike passive TV watching) may strengthen learning of aggressive behaviors through imitation.
3. **Blurred Reality:** The immersive nature of VR can blur the line between virtual and real worlds. Some young people may struggle to distinguish appropriate contexts for violence, potentially applying virtual solutions to real conflicts.
4. **Reduced Empathy:** Studies suggest that violent VR experiences may decrease empathy for victims. When violence is gamified and consequence-free in virtual environments, young people may fail to develop normal empathic responses.
5. **Potential Positive Uses:** Conversely, VR can be used for empathy training — placing users in others' perspectives to reduce prejudice and aggression. VR therapy is also being developed to treat violent tendencies.

Conclusion: While VR offers educational and entertainment benefits, unregulated exposure to violent VR content poses risks to youth development. Parental guidance, age-appropriate content ratings, and balanced usage are essential.

16. Discuss some of the applications of AI programs.

[5]

Answer: Applications of Artificial Intelligence Programs:

1. Healthcare:

- Medical diagnosis using image recognition (detecting tumors in X-rays, MRIs)
- Drug discovery and development through molecular modeling
- Personalized treatment recommendations based on patient history
- Robotic surgery assistance for precision operations
- Predictive analytics for epidemic outbreak forecasting

2. Finance:

- Fraud detection by analyzing transaction patterns
- Algorithmic trading that executes trades in milliseconds
- Credit scoring using alternative data sources
- Risk assessment for loans and investments
- Chatbots for customer service and financial advice

3. Transportation:

- Self-driving cars (Tesla Autopilot, Waymo)
- Traffic prediction and route optimization
- Autonomous drones for delivery and surveillance
- Predictive maintenance for aircraft and vehicles

4. Education:

- Intelligent tutoring systems that adapt to student learning pace
- Automated essay grading and feedback
- Personalized learning pathways
- Plagiarism detection tools

5. Business and E-commerce:

- Recommendation engines (Amazon, Netflix)
- Supply chain optimization
- Customer sentiment analysis from reviews and social media
- Automated recruitment and resume screening

6. Agriculture:

- Crop disease detection using drones and computer vision
- Yield prediction and precision farming
- Automated irrigation systems
- Livestock monitoring and health tracking

17. What are the actual or potential applications of multi-media in industry?

[5]

Answer: Applications of Multimedia in Industry:

1. Marketing and Advertising:

- Interactive product demonstrations using 3D animations and videos
- Rich media advertisements combining audio, video, and interactivity
- Virtual product catalogs and showrooms
- Social media campaigns with multimedia content for higher engagement

2. Training and Education:

- E-learning modules with video lectures, animations, and simulations
- Virtual reality training for hazardous jobs (firefighting, chemical handling)
- Interactive manuals and tutorials for equipment operation
- Gamified training programs for employee skill development

3. Product Design and Prototyping:

- 3D modeling and visualization of products before manufacturing
- Virtual prototypes that can be tested and modified digitally
- Architectural walkthroughs for real estate development
- CAD/CAM integration with multimedia visualization

4. Entertainment:

- Video games with immersive graphics, sound, and storytelling
- Streaming services for movies, music, and live events
- Theme park attractions using multimedia projections and effects

5. Healthcare:

- Medical imaging combining various scan types into 3D visualizations
- Surgical planning using multimedia reconstructions of patient anatomy
- Patient education through animated explanations of conditions and treatments

6. Communication and Collaboration:

- Video conferencing with screen sharing and collaborative whiteboards
- Digital signage for corporate communications
- Multimedia presentations for stakeholder meetings

Group C

Attempt any TWO questions. [2×10=20]

18. Write a letter of application for a job advertised in a national daily newspaper. Also, write your resume as a supporting document for the application.

[10]

Answer: Job Application Letter:

[Your Name]
[Your Address]
[City, Postal Code]
[Email] | [Phone Number]
[Date]

The Hiring Manager
[Company Name]
[Company Address]

Subject: Application for the position of [Job Title]

Dear Sir/Madam,

I am writing to express my keen interest in the position of [Job Title] advertised in [Newspaper Name] on [Date]. With my educational background in [relevant field] and [X years] of experience in [relevant area], I am confident that I possess the skills and qualifications necessary to make a valuable contribution to your esteemed organization.

During my academic career, I developed strong foundations in [relevant skills]. My previous role at [Previous Company] provided me with hands-on experience in [specific responsibilities], where I successfully [mention an achievement with metrics]. I am proficient in [technical skills] and have demonstrated ability in [soft skills like teamwork, communication, problem-solving].

I am particularly drawn to [Company Name] because of [specific reason — reputation, mission, projects]. I believe my background aligns well with your requirements, and I am eager to bring my dedication and expertise to your team.

I have attached my resume for your review and would welcome the opportunity to discuss how my background matches your needs. Thank you for considering my application. I look forward to hearing from you.

Yours sincerely,
[Your Signature]
[Your Name]

Resume:

PERSONAL INFORMATION

Name: [Your Full Name]
Address: [Your Address]
Contact: [Phone] | [Email]
Date of Birth: [DD/MM/YYYY]

OBJECTIVE

To obtain a challenging position as [Job Title] where I can utilize my skills in [relevant areas] to contribute to organizational success while advancing my professional growth.

EDUCATION

- Bachelor in Computer Applications (BCA) — [University Name], [Year]
— CGPA: [X.XX]
- +2 in Science — [College Name], [Year]
— Percentage: [XX%]

TECHNICAL SKILLS

- Programming: Python, Java, C, JavaScript
- Web Development: HTML, CSS, React, Node.js
- Database: MySQL, MongoDB
- Tools: Git, VS Code, Figma

WORK EXPERIENCE

[Job Title] | [Company Name] | [Dates]

- [Responsibility/achievement 1]
- [Responsibility/achievement 2]
- [Responsibility/achievement 3]

PROJECTS

- [Project Name] — [Brief description and technologies used]
- [Project Name] — [Brief description and technologies used]

CERTIFICATIONS

- [Certification Name] — [Issuing Organization], [Year]

LANGUAGES

- Nepali (Native)
- English (Fluent)
- Hindi (Conversational)

REFERENCES

Available upon request.

19. Suppose you represent the National Council for Educational Technology, and your boss has asked you to make comprehensive notes about the work of the Council. Explain in detail the works of the council.

[10]

Answer: Comprehensive Notes on the National Council for Educational Technology (NCET):

Introduction:

The National Council for Educational Technology is a government body established to promote the integration of technology in education, enhance digital literacy, and ensure equitable access to quality educational resources across the nation.

Key Functions and Works of the Council:

1. Policy Formulation:

The Council develops national policies and guidelines for the use of technology in educational institutions. This includes standards for ICT infrastructure, digital curriculum integration, and online assessment protocols. The policies ensure consistency and quality across public and private educational sectors.

2. ICT Infrastructure Development:

- Establishing computer labs in schools and colleges, especially in rural and underserved areas
- Ensuring reliable internet connectivity (broadband, Wi-Fi) in educational institutions
- Providing hardware (computers, projectors, tablets) to schools with limited resources
- Setting up educational data centers and cloud infrastructure for digital resources

3. Digital Content Development:

- Creating e-books, video lectures, and interactive simulations aligned with the national curriculum
- Developing open educational resources (OER) accessible to all students free of charge
- Translating quality educational content into local languages
- Establishing a national digital library with academic journals, research papers, and reference materials

4. Teacher Training and Capacity Building:

- Conducting workshops on ICT integration in teaching (using smart boards, LMS, educational apps)
- Training teachers to create digital content and conduct online classes
- Developing certification programs for educational technology specialists
- Establishing communities of practice for teachers to share best practices

5. Research and Innovation:

- Funding research on the effectiveness of educational technology interventions
- Piloting emerging technologies (AI tutors, VR labs, adaptive learning systems) in select schools
- Collaborating with universities and tech companies on EdTech innovation
- Monitoring global trends and adapting successful models to local contexts

6. Quality Assurance:

- Evaluating educational software and apps for pedagogical soundness and data security
- Setting standards for online courses and distance learning programs
- Monitoring the impact of technology integration on learning outcomes

7. Equity and Inclusion:

- Ensuring students with disabilities have access to assistive technologies

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- Bridging the digital divide between urban and rural schools
 - Providing subsidized devices and internet to economically disadvantaged students

20. Write a summary of the text "Computer Network".

[10]

Answer: Summary of "Computer Network":

A computer network is an interconnected collection of autonomous computers that can exchange data and share resources. Networks have become essential infrastructure for modern communication, business, education, and entertainment.

Types of Networks:

Networks are classified by their geographical scope. A **LAN (Local Area Network)** connects devices within a limited area such as a home, school, or office building. A **WAN (Wide Area Network)** spans large geographical areas, connecting cities, countries, or even continents — the Internet being the largest example. **MAN (Metropolitan Area Network)** covers a city or large campus, while **PAN (Personal Area Network)** connects personal devices within an individual's workspace.

Network Topologies:

The arrangement of devices in a network is called topology. Common topologies include **bus** (all devices share a single cable), **star** (all devices connect to a central hub or switch), **ring** (devices form a circular data path), **mesh** (every device connects to every other device), and **hybrid** (combinations of basic topologies). Each topology has advantages in terms of cost, reliability, and scalability.

Network Devices and Protocols:

Networks rely on specialized hardware including routers (which direct traffic between networks), switches (which connect devices within a LAN), modems (which convert digital signals to analog for transmission), and firewalls (which protect against unauthorized access). Communication follows standardized **protocols** such as TCP/IP, HTTP, FTP, and SMTP that define how data is formatted, transmitted, received, and acknowledged.

Benefits and Challenges:

Computer networks enable resource sharing (printers, files, internet connections), communication (email, video conferencing), centralized data management, and collaborative work. However, they also introduce security risks, require ongoing maintenance, and can create dependency that makes operations vulnerable to network failures.

Conclusion:

Computer networks are fundamental to the digital age, transforming how individuals and organizations interact, collaborate, and conduct business. Understanding network principles is essential for anyone working in information technology.

Note: These answers are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.